UGANDA REVISION HUB

S3 Starter Study Notes

Batch 1 • Online and offline revision

S3 • Mathematics

Linear equations in one unknown

Learning objectives

- Recognise a linear equation
- Solve equations using inverse operations
- Check a solution by substitution

Lesson notes

An equation states that two expressions are equal. A linear equation in one unknown has a variable whose highest power is 1, such as 3x + 5 = 20.

Keep the equation balanced by performing the same operation on both sides. Use inverse operations: addition is undone by subtraction, and multiplication is undone by division.

After solving, substitute your answer into the original equation to check that both sides are equal.

Worked example

3x + 5 = 20. Subtract 5 from both sides: 3x = 15. Divide by 3: x = 5. Check: 3(5)+5 = 20.

Revision questions

1 Solve 2x + 7 = 19.

2 Solve 5y - 4 = 21.

3 Check whether x = 4 solves 3x + 2 = 14.

Suggested answers

1 x = 6

2 y = 5

3 Yes; 3(4)+2 = 14

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S3 • Chemistry

Acids, bases and indicators

Learning objectives

- Describe common properties of acids and bases
- Explain the use of indicators
- Interpret the pH scale

Lesson notes

Acids in water produce hydrogen ions, while alkalis are soluble bases that produce hydroxide ions in water. Laboratory chemicals must never be tasted or touched to identify them.

Indicators change colour depending on whether a solution is acidic, neutral or alkaline. Blue litmus turns red in acid; red litmus turns blue in alkali. Universal indicator gives a range of colours linked to pH.

The pH scale is commonly shown from 0 to 14: values below 7 are acidic, 7 is neutral, and values above 7 are alkaline. Stronger acids have lower pH values.

Worked example

A solution with pH 3 is acidic; a solution with pH 7 is neutral.

Revision questions

1 What colour does blue litmus turn in acid?

2 Classify pH 11 as acidic, neutral or alkaline.

3 Why should learners never taste laboratory chemicals?

Suggested answers

1 Red

2 Alkaline

3 They may be poisonous, corrosive or otherwise harmful.

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